

# GIHUN KIM

Ph.D. in Engineering

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## EDUCATION

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- Yonsei University**, South Korea *Mar. 2015 - Feb. 2024*  
Ph.D., School of Integrated Technology, College of Engineering  
Thesis: Implementation of CNN-based ideal model observer for breast CT image  
(advisor: Prof. Jongduk Baek)
- Yonsei university**, South Korea *Mar. 2012 - Feb. 2015*  
B.S., School of Integrated Technology, College of Engineering
- Korea Science Academy of KAIST**, South Korea *Mar. 2009 - Feb. 2012*

## RESEARCH INTERESTS

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I am interested in Deep-learning-based application for computer vision.  
My research during Ph.D. education includes application of deep learning techniques appropriately for medical images.

- Quality and performance evaluation of medical imaging (observer study)
- Signal segmentation and Computer-Aided Diagnosis in CT images
- Synthetic CT image generation using Generative model
- Denoising and artifact reduction of CT images (e.g., Low-dose CT, Metal artifact, Cone-beam artifact)

Currently, I've touched general deep-learning-based application for computer vision, including belows:

- Single image super resolution
- Multi-exposure image fusion
- De-mosaic and de-noise for raw image from image sensor
- Noise estimation and rendering

Also, during working experience, I've involved below: - Image restoration using conventional computer vision methods

- Camera exposure and color temperature adjusting
- On-device AI model analysis and optimization

## PUBLICATIONS

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**G. Kim**, M. Han, H. Shim, and J. Baek, A convolutional neural network-based model observer for breast CT images, *Medical Physics*, 2020.

**G. Kim**, M. Han, and J. Baek, Strategy to implement a convolutional neural network based ideal model observer via transfer learning for multi-slice breast CT images, *Physics in Medicine and Biology*, 2023.

**G. Kim** and J. Baek, Power-law spectrum-based objective function to train a generative adversarial network with transfer learning for the synthetic breast CT image, *Physics in Medicine and Biology*, 2023.

## CONFERENCE

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**G. Kim**, C. Lee, H. Nam, and J. Baek, A new method to improve lesion segmentation accuracy for breast computed tomography images, poster presented at *IEEE International symposium Biomedical Imaging*, 2017.

**G. Kim**, H. Shim, and J. Baek, Feasibility study of deep convolutional generative adversarial networks to generate mammography images, poster presented at *SPIE medical imaging* and published in *Proceeding*, 2018.

**G. Kim**, M. Han, H. Shim, and J. Baek, Implementation of an ideal observer model using convolutional neural network for breast CT images, oral presented at *SPIE medical imaging* and published in Proceeding, 2019.

**G. Kim** and J. Baek, Implementation of CNN-based multi-slice model observer for 3D cone beam CT, oral presented at *SPIE medical imaging* and published in Proceeding, 2021.

**G. Kim** and J. Baek, Feasibility study of using masked auto-encoder for a streak artifact reduction in sparse view CT, poster presentation at *SPIE medical imaging* and will be published in Proceeding, 2024.

## PATENTS

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J. Baek, S. Choi, **G. Kim**, M. Han, PET-RFA complex medical device and treatment method using the same, *US Patent, US10105111B2*, issued Oct. 23, 2018.

J. Baek, S. Choi, **G. Kim**, M. Han, PET-RFA COMPLEX MEDICAL DEVICE AND METHOD FOR TREATING USING IT, *Korean Patent, 10-146252*, issued Nov. 12, 2014.

J. Baek, **G. Kim**, J. Kim, System and Method for Dot counting projection for fast calculation, *Korean Patent, 10-1691654*, issued Dec. 26, 2016.

## AWARDS AND SCHOLARSHIP

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Full-funded Scholarship during undergraduate course and graduate coursework.

Awarded the 2014 Creative ICT Convergence Korea Forum Excellence award for Creative Task.

Nominated the 2021 Robert F. Wagner All-Conference Best Student Paper in SPIE Medical imaging 2021.

## RESEARCH DURING DOCTORAL COURSE

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### **Image quality assessment using observer study.**

Model observers have been developed to evaluate the lesion detection performances for medical imaging systems. Among such model observers, the ideal model observer provides an upper bound of the detection performance, but it is difficult to implement using conventional linear functions. I proposed a convolutional neural network-based efficient model observer for breast CT images. The proposed model observer provided higher detection performance than the conventional linear model observers. Also, I extended this work to the 3D CT images (i.e., multi-slice images). Proposed CNN-based model observers also provided higher detection performance in the multi-slice images, and I reduced the required number of training dataset using transfer learning.

Knowledge: Image quality assessment, breast CT, observer study, ideal observer, classification problem

> *Thesis*, 2 journal article (*MP2020, PMB2023*), 3 conference presentations (*ISBI2017, SPIE MI 2019, SPIE MI 2021*)

### **Synthetic breast image generation using GAN**

Breast phantoms have been widely used in the medical imaging field for various purposes, such as deep-learning model training, image quality assessments and radiology training tools. The advances in deep learning have improved the importance of quantified breast image datasets with accurate labels for network training. I proposed a new objective function to train the GAN for the breast CT images. With the proposed objective function, the GAN provided the lowest FID score and achieved higher quality. I also analyzed the effects of transfer learning datasets from different image domains for the GAN.

Knowledge: GANs, objective function, breast CT images, synthetic breast phantom,

Power-law spectrum in breast images.

> 1 journal article (*PMB2023*), 1 conference presentations (SPIE MI 2018)

### **Streak artifact reduction in sparse view CT using masked auto-encoder**

Since CT uses X-rays, the dose exposure of patients has been raised as a major side-effect. While the dose exposure could increase the incidence of cancer, researchers have been trying to reduce the dose of CT scan. I employed masked auto-encoder (MAE) for sinogram inpainting to reduce the streak artifact in sparse view CT, taking their successful application in image inpainting tasks. Also, I exploring the additional down-stream task using encoder module of MAE for observer study or lesion segmentation.

Sparse view CT, image inpainting, masked auto-encoder.

> 1 conference presentations (SPIE MI 2024)

## TECHNICAL SKILLS

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|-------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Language                | Korean (native), English (proficient), Japanese (beginner)                                                              |
| Programming             | Python , C/C++(Mostly used), MATLAB, Kotlin, Java, JavaScript (particularly Nodejs and React.js)                        |
| Software & Tools        | DL frameworks (PyTorch, TensorFlow, Tensorboard, Keras), DL model pools (HuggingFace, TIMM, ollma), Numpy, Scikit-learn |
| Development Environment | Ubuntu, Docker, Kubernetes, Singularity                                                                                 |

## WORK EXPERIENCE

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**Medisys Lab., Department of Artificial Intelligence, Yonsei university** *August 2023 - June 2024*  
*Senior researcher*

I worked in Medisys Lab. as senior researcher and post-doctoral researcher. The main research topic is streak artifact reduction for sparse view CT using representation learning (especially Masked auto-encoder), and exploring down-stream task extension. Also, interested in observer study and metal artifact reduction.

**Visual Solution team, Mobile Experience division, Samsung electronics Co.** *July 2024 - current*  
*Staff engineer*

Currently, working as staff engineer in MX (Mobile eXperience) division, Samsung electronics Co. The main task is enhancing the image quality in zoom scenario of camera system, expecially targetting Samsung Galaxy flagship series. My target is visual image restoration using single-image/multi-frame super resolution, multi-exposure fusion for HDR, detail enhancement during post-processing and noise rendering in zoom scenario. With my target, I also have jobs in analysis for remosaic/demosaic algorithm using AI. While my work has been for On-device solution, I've also involved with On-device model analysis and optimization.

## PERSONAL TRAITS

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- Highly motivated and eager to learn new things.
- Co-operative with co-workers and proficient in team-level project.
- Ability to work as an individual as well as in group.
- Highly interesting in AI-based development tools (such as LLM-based supporting tools)